

GRANTS-MILAN, NM, USA

WMO: 723625

Lat: 35.165N

Lon: 107.902W

Elev: 6520

StdP: 11.55

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 93057

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.5	8.5	-9.4	4.2	25.7	-5.7	5.2	26.5	26.5	43.0	24.3	41.4	6.1	310	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.4	91.5	57.3	89.6	56.4	87.4	55.8	62.4	77.6	61.3	76.5	60.3	75.5	10.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	94.6	64.7	57.3	89.7	64.3	56.0	85.3	64.1	31.4	76.8	30.5	76.1	29.7	75.5	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.3	21.7	19.1	-5.5	95.9	6.8	2.1	-10.4	97.4	-14.4	98.7	-18.3	99.9	-23.2	101.4		
			-6.1	64.7	6.4	1.0	-10.7	65.4	-14.4	66.0	-18.0	66.6	-22.6	67.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	30.8	34.6	41.8	48.6	56.5	67.6	71.6	68.6	61.7	50.9	39.7	31.4
	DBStd	15.53	7.51	7.02	6.55	6.32	6.60	4.96	3.32	3.16	4.97	6.30	7.23	8.37
	HDD50	2363	594	431	263	98	21	0	0	0	2	64	313	577
	HDD65	5783	1059	851	720	492	268	30	1	6	118	437	759	1042
	CDD50	2512	0	0	8	56	224	529	669	576	354	92	4	0
	CDD65	456	0	0	0	0	6	108	205	117	20	0	0	0
	CDH74	8656	0	0	3	31	574	2689	2728	1819	739	73	0	0
	CDH80	3306	0	0	0	0	121	1171	1194	663	153	4	0	0
	WSAvg	8.1	7.8	8.4	9.1	10.1	9.3	8.6	7.4	6.4	7.1	7.6	7.9	7.5
	PrecAvg	9.8	0.4	0.4	0.4	0.6	0.5	0.5	1.7	2.2	0.9	0.8	0.6	0.6
Precipitation	PrecMax	15.7	1.8	1.5	2.1	2.5	2.7	1.7	4.0	4.8	2.9	6.3	2.6	2.6
	PrecMin	5.5	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.9	0.0	0.0	0.0	0.0
	PrecStd	2.6	0.5	0.4	0.6	0.7	0.6	0.5	0.9	1.0	0.7	1.3	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.5	66.4	73.6	77.4	87.6	94.4	94.9	91.5	86.8	79.5	70.5	61.8
		MCWB	40.4	43.3	45.9	47.7	53.5	55.7	58.4	59.6	56.2	50.8	45.6	40.5
	2%	DB	54.8	61.7	69.8	74.2	83.2	91.5	92.0	89.2	83.9	75.6	66.3	56.9
		MCWB	38.8	41.0	44.4	46.5	51.0	55.4	58.1	57.9	55.2	49.6	44.2	39.6
	5%	DB	51.4	56.9	66.0	71.9	80.4	89.6	89.7	86.8	81.6	72.9	62.8	52.3
		MCWB	37.0	38.8	42.6	45.6	50.1	54.5	57.6	57.4	54.2	48.8	42.8	37.5
	10%	DB	46.8	52.0	62.1	68.9	76.9	87.2	87.1	84.0	78.9	69.8	57.6	47.4
		MCWB	35.0	37.1	41.1	44.3	48.5	53.4	57.5	57.7	53.8	48.1	40.8	36.1
	0.4%	WB	44.0	44.9	47.0	50.5	56.5	61.3	64.4	64.3	60.9	57.1	47.7	44.1
		MCDB	54.2	61.4	69.6	71.3	80.1	85.1	81.3	80.0	73.2	66.0	62.8	54.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.4	42.1	45.2	48.2	53.3	59.0	62.5	62.4	59.4	53.8	45.8	41.3
		MCDB	51.0	57.4	66.8	69.5	75.9	80.0	78.2	76.4	71.1	66.3	61.1	53.1
	5%	WB	38.0	40.1	43.5	46.4	51.5	57.2	61.5	61.4	58.1	51.5	44.1	39.0
		MCDB	48.3	54.3	63.2	68.2	73.0	79.8	76.8	75.9	70.7	65.9	59.7	49.8
	10%	WB	36.0	37.9	41.8	44.9	49.9	55.7	60.5	60.4	56.8	49.6	42.1	36.6
		MCDB	45.4	50.4	59.9	66.5	72.5	79.4	76.3	74.6	70.4	65.2	56.6	46.9
Mean Daily Temperature Range	5% DB	MDBR	29.6	30.5	34.6	34.4	35.2	37.1	30.4	30.5	32.0	33.9	33.2	28.8
		MCDBR	36.8	41.5	43.0	40.5	40.2	40.3	35.7	36.8	37.3	39.9	40.5	36.3
		MCWBR	22.5	23.3	22.3	20.3	18.4	17.0	12.1	13.1	14.7	18.8	22.2	21.5
		MCDBR	30.8	35.6	38.9	36.1	35.2	33.5	27.6	27.1	26.5	31.1	35.2	31.8
	5% WB	MCWBR	19.2	20.3	20.4	18.4	16.5	13.7	8.9	9.6	10.3	15.5	19.9	19.4
Clear-Sky Solar Irradiance	taub		0.233	0.238	0.260	0.279	0.291	0.310	0.354	0.333	0.296	0.270	0.244	0.235
	taud		2.658	2.655	2.580	2.509	2.524	2.451	2.424	2.448	2.564	2.603	2.631	2.652
	Ebn at Noon		319	327	324	319	314	306	292	298	307	309	311	312
	Edh at Noon		22	25	29	33	33	36	36	35	30	26	22	21
All-Sky Solar Radiation	RadAvg		1020	1323	1778	2175	2430	2541	2201	2015	1809	1486	1134	899
	RadStd		85	108	111	115	157	129	113	82	70	115	69	58

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+59

Nomenclature: See separate page